# Effectiveness of a Treatment for Childhood Obesity: A Case-Based Approach

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**Abstract**

**Introduction**: Childhood obesity is a global public health concern, with increasing prevalence and long-term severe implications. Understanding the effectiveness of standardized treatment for childhood obesity and identifying predictive factors of treatment success is crucial for developing personalized intervention strategies.

**Method**: This study evaluates the results of standardized treatment for childhood obesity using regression techniques to predict outcomes in new patients. With the help of the Elx-Crevillent Health Department, patient data was obtained. Different linear regression models were applied to predict treatment success.

**Results**: The analysis showed that initial BMI-SD, the time between visits, and family history of obesity significantly influence treatment outcomes. The Lasso and Ridge models provided superior predictive accuracy compared to traditional methods.

**Discussion**: These findings allowed the development of patient profiles with a higher probability of success from two different perspectives, depending on how treatment success is defined.

*Keywords:* childhood obesity, successful treatment, regression model, decision making

## 1. Introduction

Childhood obesity has emerged as a critical global public health issue, its prevalence increasing at an alarming rate in recent decades [1]. This epidemic poses significant long-term health risks, including an increased likelihood of developing chronic conditions such as type 2 diabetes, cardiovascular diseases, and increased risk of premature mortality [2]. The escalating incidence of childhood obesity requires effective intervention strategies that not only address the condition but also prevent its onset and progression.

Multiple treatment programs have been developed and implemented to tackle childhood obesity. However, the variability of individual responses to these interventions highlights the need for a broader understanding of the efficacy of treatment. Identifying which factors contribute to successful results can significantly improve the personalization of treatment plans, improving the probability of achieving and maintaining healthy weight among pediatric patients[3, 4, 5].

This study aims to evaluate the effectiveness of a standardized treatment program for childhood obesity implemented in the Pediatric Service of the Elx-Crevillent Health Department. By applying various regression techniques, including linear regression, multiple linear regression, and Lasso and Ridge models, this research seeks to predict treatment outcomes and identify key predictive factors. These advanced statistical methods are used to improve the accuracy of predictions compared to traditional approaches.

This research's primary objective is to assess the intervention program's overall success and develop predictive profiles that can help tailor treatments to individual patient characteristics. Success in this context is defined by a significant reduction in the age- and gender-adjusted Body Mass Index standard deviation (BMI-SD), a widely used metric for evaluating obesity in children.

The results of this study are expected to provide valuable information on the determining factors for the success of childhood obesity treatment. By understanding how sociodemographic, genetic, neonatal, lifestyle, clinical, and treatment intervention factors influence treatment outcomes, healthcare professionals can better design and implement personalized intervention strategies. In the end, this research aims to increase the efficacy of obesity treatment programs, thus improving the health and well-being of children affected by this global obesity problem.

The article follows a structured format: Section 1, the introduction provides the study's context and establishes its objectives. Section 2 focuses on study design, detailing the data preparation and study design techniques. Section 3 shows the ongoing research on this topic. Section 4 describes the methodology, including the regression method employed and the categorization of the results. The study findings are presented in Section 5, where the results obtained are presented. In Section 6, a discussion is introduced on the results of categorizing the outcome of the treatment. Finally, in Section 7, the findings are summarized, highlighting the implications of the results and the potential of artificial intelligence in treating childhood obesity.

## 2. Literature Review

The increase in the prevalence of obesity worldwide is becoming a worldwide problem [6, 7]. In 2013, more than 42 million children under five years of age were overweight or obese, and developing countries experienced a higher rate of increase [7]. The rise in childhood obesity is attributed to a variety of factors, including poor eating habits, such as increased consumption of high-energy processed foods and decreased intake of fruits and vegetables, which contribute significantly to this problem in addition to decreased physical activity and genetic predisposition [8, 9]. Reduced physical activity and increased sedentary behaviors, particularly screen time, are also major contributors [8, 10, 11]. This trend has significant health implications, including psychological disorders and increased risk of non-communicable diseases later in life, like dyslipidemia, hypertension, diabetes mellitus, non-alcoholic fatty liver disease, and obstructive sleep apnea [6, 12]. The psychological and social consequences of childhood obesity are often overlooked but prevalent [13].

Socioeconomic factors play a significant role in the prevalence of childhood obesity. Lower socioeconomic status is often associated with higher rates of obesity due to limited access to nutritious foods, fewer opportunities for physical activity, and lower health literacy. Additionally, disparities in obesity rates are evident among different racial and ethnic groups, with minority populations often experiencing higher rates of obesity and related health issues [14, 15]. Addressing these disparities requires targeted interventions that consider the unique challenges faced by these communities. Baseline BMI and waist-hip ratio are essential determinants of weight reduction success[16]. Multidisciplinary, family-focused interventions incorporating dietary, psychosocial, and endocrine counseling show promise in treating childhood obesity [16, 17].

These health issues can persist into adulthood, increasing the likelihood of early morbidity and mortality [13]. Prevention is crucial, with supportive policies, environments, and communities playing a fundamental role in shaping healthier choices for children and parents [7]. Early intervention is crucial, as childhood obesity often leads to adult obesity and related health issues [13, 18]. Improving dietary intake and increasing physical activity remain the best therapeutic approaches to mitigate obesity-related risks in children [18]. Management strategies mainly focus on lifestyle modifications, dietary changes, and increased physical activity, with pharmacotherapy and bariatric surgery reserved for severe adolescent cases [19]. Addressing this epidemic requires coordinated efforts from governments, civil society, and other stakeholders[6].

## 3. Study design

This section explains the design of this case study, including an overview of the database, its structure, the data collection process, and the limitations encountered. The study population comprised 96 children aged 5 to 15 who participated in an obesity intervention program at the Pediatrics Service of the Elx-Crevillent Health Department.

### *3.1. Database*

For the development of this study, careful ethical measures were adopted in data collection, ensuring confidentiality and anonymity of patient information and obtaining informed consent from participants under the guidelines established by regulatory authorities and medical research ethics committees. A gender perspective was incorporated into the data analysis, ensuring equal inclusion of boys and girls and avoiding stereotypes. The study was conducted with a commitment to contribute to the well-being of participants and society, which aligns with the ethical principles of medical research.

In addition to these ethical considerations, several strategic methods for data collection were applied. A comprehensive literature review involving scientific studies and international databases was conducted. This review provided a thorough understanding of the main medical challenges and methodologies required to use machine learning algorithms for predicting childhood obesity, thus establishing a solid basis for the study design. Furthermore, collecting the data presented significant challenges due to the sensitive nature of medical information involving minors. This process required obtaining consent, ensuring privacy, and maintaining confidentiality. Data from minors were collected through a questionnaire given to parents at the time of the initial visit and after the completion of the treatment. This questionnaire allowed us to obtain the variables detailed in the following subsection 3.2. Additionally, the aspect of the study necessitated careful and consistent tracking of the participants over an extended period, with an average time of approximately 7.5 months between the initial visit and subsequent follow-ups, which added further complexity to the data collection process.

The data quality was checked, and there were no missing values. In addition, categorical variables were transformed into numerical ones; however, the nature of these data was considered when modeling and interpreting the results.

### *3.2. Variables*

The variables selected for this study were chosen based on their potential impact on childhood obesity, guided by extensive reviews of the literature and expert consultations. These variables include sociodemographic, genetic, neonatal, lifestyle, clinical, and treatment intervention factors.

To understand the socioeconomic and cultural context that can affect obesity rates, factors such as sex, age, and income provide an essential context. For example, family status and income analysis help identify socioeconomic disparities that may influence children's access to healthy foods and recreational activities. In addition, lifestyle choices, including diet, physical activity, and screen time, are modifiable factors that can significantly influence childhood obesity and provide a global view of daily habits. Adherence to the Mediterranean diet was assessed through the KIDMED score (range - 4 to 12)[20].

Genetic predispositions also play a key role in the development of obesity. Variables related to parental height, weight, and BMI allow for a better understanding of the heritable aspects of obesity. By including these genetic factors, it is possible to assess the genetic impact on childhood obesity. Similarly, neonatal variables, such as gestational age, birth weight, and breastfeeding status, provide early health indicators. These variables help to understand how early life conditions influence long-term health outcomes and can serve as a basis for early obesity prevention interventions.

Finally, in terms of clinical measurements taken at the initial visit, including BMI, abdominal circumference, and personal and family medical history, they establish a baseline to assess the health status of the children. These measurements are essential to track changes over time and assess the effectiveness of the intervention. In addition, treatment intervention variables, such as the duration between visits and changes in BMI, provide direct indicators of treatment success. By monitoring these variables, health professionals can identify the most effective interventions and make the necessary adjustments to improve outcomes.

Table 1 summarises the characteristics of the dataset and its types, providing an overview of the variables included in the study. These variables were collected through a questionnaire provided to the patient's parents at the initial visit.

### *3.3. Descriptive analysis*

Once the database and variables are explained, the next step is the descriptive statistical analysis of the preparatory action focused on childhood obesity in the Elx-Crevillent Health Department. This analysis is essential to understanding variables' distribution and key characteristics, identifying possible risk factors and predictor variables for success in intervention programs, and preparing the way for developing effective interventions and predictive models.

To begin with, socioeconomic and neonatal variables were explored in order to know if there is an influence on treatment outcomes for this study case. As mentioned

Table 1: Dataset features.

| Feature | Description | Type |
|---|---|---|
| **Socio-demographic variables** | | |
| sex | Sex of patient | Categorical |
| age | Age of patient when treatment starts (t_visit0) | Numeric |
| family_status | Indicates family situation | Categorical |
| n_siblings | Number of siblings, including the patient | Numeric |
| postal_code | Postal code | Categorical |
| birth_country | Birth country of the patient | Categorical |
| income | Classification into groups depending on income | Categorical |
| birth_country_mother | Mother's country of birth | Categorical |
| study_mother | Mother's level of education | Categorical |
| job_mother | Mother's employment status | Categorical |
| birth_country_father | Father's country of birth | Categorical |
| study_father | Father's level of education | Categorical |
| job_father | Father's employment status | Categorical |
| **Genetics variables** | | |
| height_mother | Mother's height measured in centimeters (cm) | Numeric |
| weight_mother | Mother's weight measured in kilograms (kg) | Numeric |
| BMI_mother | Mother's body mass index (BMI) | Numeric |
| height_father | Father's height measured in centimeters (cm) | Numeric |
| weight_father | Father's weight measured in kilograms | Numeric |
| BMI_father | Father's body mass index | Numeric |
| **Neonatal variables** | | |
| gestational_age | Classification according to the length of gestation during pregnancy | Numeric |
| weight_birth | Patient's weight at birth in grams (g) | Numeric |
| breastfeeding | If the patient was breastfed or not | Bool |
| breastfeeding_exc | In case of breastfeeding, if it was exclusive or not | Bool |
| t_breastfeeding | Duration in months of (non-)exclusive breastfeeding received by the patient | Numeric |
| **Lifestyle variables** | | |
| KIDMED | Assesses patients' adherence to the Mediterranean diet | Categorical |
| rest | Average daily hours sleep duration | Numeric |
| breakfast | If the patient eats breakfast or not | Bool |
| sport | Average weekly hours sport duration | Numeric |
| t_screen | Average weekly hours of exposure to electronic devices with displays | Numeric |
| **Clinical variables** | | |
| t_visit0 | Patient's age in months, at the time of the initial visit | Numeric |
| weight_0 | Patient's weight, measured in kg, at the time of the initial visit | Numeric |
| size_0 | Patient's height, measured in cm, at the time of the initial visit | Numeric |
| BMI_0 | BMI of the patient, expressed in kg/m$^2$, at the time of the initial visit | Numeric |
| BMI-SD_0 | Difference between actual and expected BMI for their age and sex at the initial visit | Numeric |
| hip_girth | Patient's hip girth, measured in cm, at the time of the initial visit | Numeric |
| abdominal_girth | Patient's abdominal girth, measured in cm, at the time of the initial visit | Numeric |
| personal_history | If there is a personal history of obesity-related diseases in a patient | Bool |
| family_history | If there is a familiar history of obesity-related diseases in a patient | Bool |
| **Treatment intervention variables** | | |
| t_v0_v1 | Duration in months between initial and first visit after starting obesity treatment | Numeric |
| weight_1 | Patient's weight, measured in kg, at the first visit | Numeric |
| size_1 | Patient's height, measured in cm, at the first visit | Numeric |
| BMI_1 | BMI of the patient, expressed in kg/m$^2$, at the first visit | Numeric |
| BMI-SD_1 | Difference between actual and expected BMI for their age and sex at the first visit | Numeric |
| result | Change in BMI, calculated as the difference between BMI-SD_0 and BMI-SD_1 | Numeric |

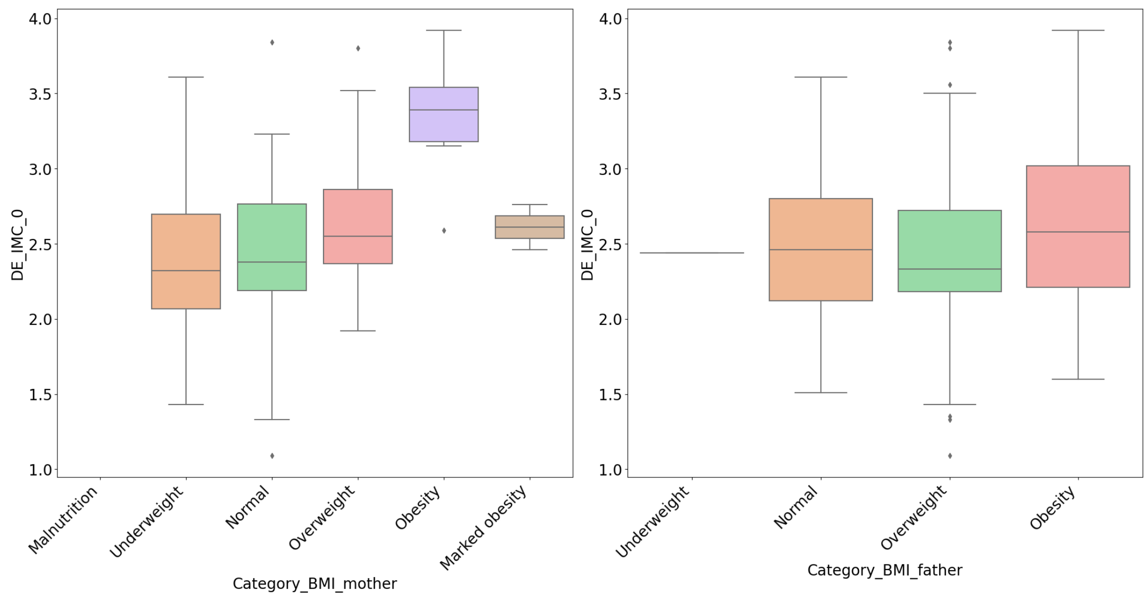


Figure 1: Analysis of parental BMI relative to the child's BMI-SD at baseline.

in section 3.2, these include variables related to parents and patients such as gender, age, family status, number of siblings, weight, birth, and breastfeeding, among others. Despite not finding highly significant correlations between these variables and treatment outcomes, some critical insights emerged.

Figure 1 presents the parents' BMI analysis concerning the children's BMI at the initial visit. The data reveal a noticeable trend: children's BMI tends to increase proportionally with their mothers' BMI, particularly in cases of obesity. This correlation is less evident with the fathers. These findings suggest a significant maternal influence on childhood obesity, possibly due to genetic factors.

When examining lifestyle factors, Figure 2 shows the progression of screen time, sports activity, and sleep duration by age. Screen time increases significantly with age, while sleep duration decreases slightly. Physical activity, measured in weekly sports hours, remains relatively constant, ranging from 2 to 5 hours. The World Health Organization (WHO) recommends that children aged 6 to 12 years sleep 9 to 12 hours a night, while adolescents should aim for 8 to 10 hours. However, a study by the *Instituto de Investigaciones del Sueño* [21] indicates that up to 60 % of Spanish children do not meet these recommendations, and one in three shows signs of fatigue and sleepiness during the day. These findings underscore the need to address excessive screen time and insufficient sleep as part of obesity prevention efforts. The increase in screen time with age is particularly concerning, as it could

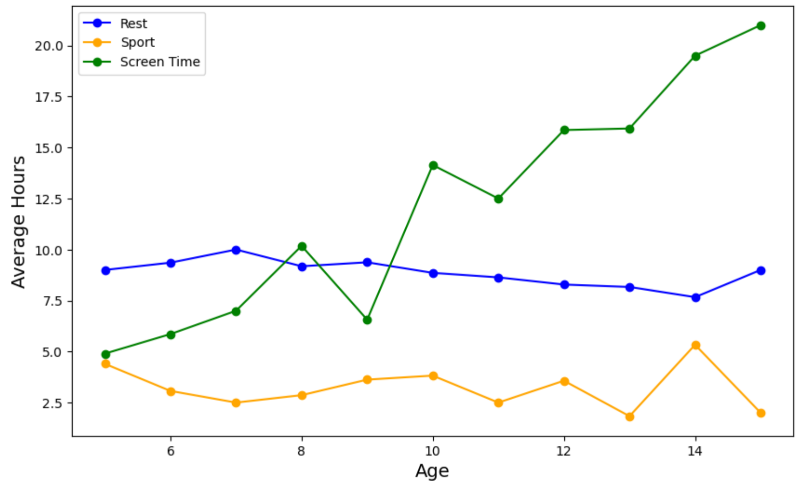


Figure 2: Evolution in hours of screen use, sport, and rest according to age.

contribute significantly to the rise in obesity rates.

This is further connected to the findings in Figure 3, where a substantial proportion of children suffer from overweight or obesity. Data also differentiate between boys and girls, showing a higher prevalence of overweight in boys. This information is essential to identify the demographic characteristics of the at-risk population and target interventions accordingly. The gender differences in the BMI categories suggest that boys may be more susceptible to obesity-related factors, which could be related to different lifestyle or environmental influences, as seen in Figure 2.

## 4. Methodology

This section presents the various regression methods used to achieve the best fit with the initial patient data. First, the response variable in the regression models will be defined. Then, variables that can act as predictors in the model will be selected. Different regression models will be evaluated with these variables to identify the best fit. The treatment will be categorized based on the regression results to determine which patients have been successful and which have not.

### *4.1. Dependent variable*

Knowing the variable used to evaluate treatment success in new patients is essential to defining the dependent variable in the regression study. Due to the continuous growth in children, BMI is not considered a directly applicable indicator. Therefore,

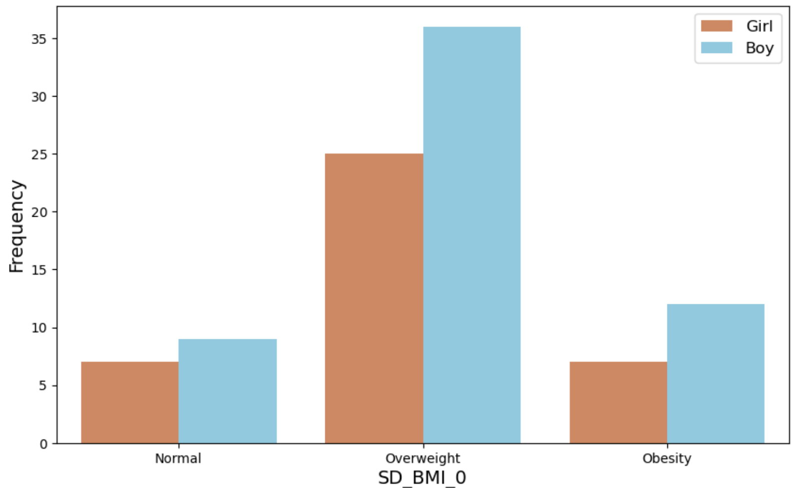


Figure 3: Analysis of the SD distribution of BMI at initial visit.

sex- and age-specific tables evaluate BMI in this demographic group. In particular, the BMI-SD is employed, which considers the difference between the observed BMI in the patient and the expected BMI for an individual of the same age and sex [22]. Calculating this difference requires information on the patient's sex, age, weight, and height. By utilizing resources such as the Spanish Society of Pediatric Gastroenterology, Hepatology, and Nutrition website, comparisons are ensured to be based on specific norms for the pediatric population in Spain, following the WHO guidelines from 2006/2007 [23].

Our study measured BMI-SD at the initial visit (0) and post-treatment visits (1). The success of treatment is evaluated by the difference between the BMI-SD measured after starting treatment and the BMI-SD recorded at the initial visit (Equation 1). A negative result suggests treatment success.

$$Result = (BMI - SD_1) - (BMI - SD_0) \tag{1}$$

Although the study on obesity treatment aims to predict treatment success through regression, the dependent variable in the regression is the standard deviation of the body mass index at the end of the treatment. This variable determines the result of the treatment (Result), and with different thresholds, it can indicate whether the outcome is successful. Using the standard deviation of the body mass index at the end of the treatment (BMI-SD_1) as the dependent variable in the regression allows a more detailed, precise, and comprehensive analysis of the effect of the independent variables on the outcome of the treatment. This provides more valuable and detailed information to evaluate and improve childhood obesity treatment programs.

### *4.2. Feature Selection*

In the set of independent variables, the most relevant features of the dataset are selected to improve interpretability and performance in predictive models. We use four techniques to achieve this, each yielding different vital features. The union of these sets provides a comprehensive set of features essential to the model, while the intersection of these sets highlights the most predictive features of our model. The techniques used for feature selection in this case are Recursive Feature Elimination (RFE)[24], ElasticNet [25], Gradient Boosting [26] and RFECV Recursive Feature Elimination with Cross-Validation with Gradient Boosting Regressor (RFECV) [27].

### *4.3. Regression Models*

Using different regression models, we model the relationship between the dependent and independent variables (s) obtained from the feature selection process. These models allow us to predict the dependent variable's value based on the independent variables' values with varying degrees of accuracy. The models used are as follows:

i) Linear Regression is a statistical analysis technique used to model the relationship between the dependent variable (target variable) and an independent variable (predictor). The objective is to find a linear equation that describes this relationship as accurately as possible, allowing the prediction of the values of the dependent variable based on the independent variable [28].

ii) Multiple Linear Regression is a statistical analysis technique used to model the relationship between a dependent variable and two or more independent variables. The objective is to find the linear equation that best fits the observed data, allowing for predicting the dependent variable's value [29].

iii) Lasso Regression (Least Absolute Shrinkage and Selection Operator) is a regression model with an L1 penalty. This penalty forces some coefficients to be exactly zero, eliminating certain irrelevant features of the model [30].

iv) Ridge Regression is a linear regression technique with an L2 penalty. This penalty adds the sum of the squares of the coefficients to the loss function. This model handles multicollinearity and reduces the risk of overfitting by shrinking the magnitude of the coefficients without forcing them to zero [31].

v) ElasticNet Regression is a linear regression technique that combines the penalties of Lasso (L1) and Ridge (L2). Combining both penalties makes this model especially useful for handling data with highly correlated features and selecting the most relevant ones [32].

vi) Polynomial Nonlinear Regression is an extension of linear regression that allows for modeling nonlinear relationships between the independent and dependent variables. Instead of using only linear terms, this model includes polynomial terms (quadratic, cubic, etc.) of independent variables, enabling the capture of curvatures and more complex patterns in the data [33].

In this study, we will use the adjusted coefficient of determination ($R^2$ adjusted) and the root mean square error (RMSE) to evaluate the regression models. The adjusted $R^2$, which accounts for the number of predictors in the model, only increases if the new variables genuinely improve the model [34].

The RMSE measures the average magnitude of the model's prediction errors and calculates the standard deviation of the differences between the predicted and observed values [35]. A lower RMSE indicates a better fit of the model to the observed data. Using these two metrics together, we will evaluate the model's accuracy and generalization ability to new data.

Once the best-fitting model is selected, it is crucial to verify that certain assumptions are met to ensure the validity and accuracy of the final model. The main assumptions are linearity, homoscedasticity (Breusch-Pagan test [36], White test [37], and Goldfeld-Quandt test [38]), independence of errors (Durbin-Watson test [39]), normality of errors (Anderson-Darling test [40]) and absence of multicollinearity (Variance Inflation Factor [41]).

Meeting these assumptions is fundamental to ensuring that the regression model is robust and reliable and that the inferences and predictions derived from the model are valid and accurate.

### *4.4. Success Rating Threshold*

The variable 'Result' is a key indicator to evaluate the effectiveness of the treatment applied in the fight against childhood obesity in each patient. Although the regression analysis uses the variable BMI-SD_1 as the dependent variable, it is the "Result" variable that will allow us to assess the success of the treatment in the patient. Therefore, the regression outcome must be used to obtain this variable.

A negative value in the "Result" variable reflects an improvement in the patient's BMI, approaching the typical value corresponding to their age, height, and weight at the follow-up visit compared to the initial visit. In other words, a negative result indicates that treatment has managed to bring the patient's condition closer to healthier BMI parameters. However, it is essential to note that not all negative results are equally significant. A value close to 0 could suggest an insignificant improvement, possibly influenced by factors other than the treatment.

Two scenarios are proposed to categorize the results and determine the importance of improving obesity treatment. The first scenario is based on the analysis conducted in the study [42] on the effects of BMI-SD reduction on cardiovascular risk factors. For our study, a threshold of -0.15 was set in the outcome variable to define the success of treatment. This threshold was explicitly adjusted for the 6-month treatment period based on the findings provided by the mentioned research. All results below this value indicate a more significant decrease in BMI-SD, representing treatment success, while higher values indicate treatment failure.

The second scenario, following the methodology used by [43], identified that changes in BMI-SD are associated with improvements in the body fat percentage of the pediatric population. In this study, four different ranges are defined to evaluate the "Result" variable: i) A value greater than -0.6 implies no significant improvement in the treatment; ii) A value between -0.6 and -1.5 indicates a significant reduction in body fat, that is, a significant improvement; iii) Values between -1.5 and -1.8 indicate an excellent treatment outcome and iv) Values below -1.8 indicate an outstanding treatment outcome.

This categorization allows for a more detailed assessment of the treatment's impact and helps identify more precise levels of success in the fight against childhood obesity. It also helps to build profiles of patients with successful treatment outcomes, which can guide specialists in decision making.

As a summary of the previous sections, figure 4 shows the steps of the process from the beginning of the research, deciding which variables are going to be collected, to the categorization of the results of the patients that lead us to obtain a profile of the patient who is successful in the treatment.

## 5. Results

After having defined in subsection 4.3 the possible models to be used to model the relationship between the dependent and independent variables, Table 2 contains a comparison of various regression models along with the number of variables each uses and various metrics.

Based on the comparison of various regression models, it is evident that multiple regression models show minimal improvement with additional predictors, as R-square increases only slightly from 0.8050 to 0.8080 when moving from one to three variables. Regularization models (ElasticNet, Lasso, and Ridge) outperform multiple regression models. ElasticNet and Lasso have an R-square of 0.8607 and RMSE around 0.2426, indicating better accuracy and lower error. Ridge stands out with the highest R-square of 0.8633 and the lowest RMSE of 0.240270, making it the most precise

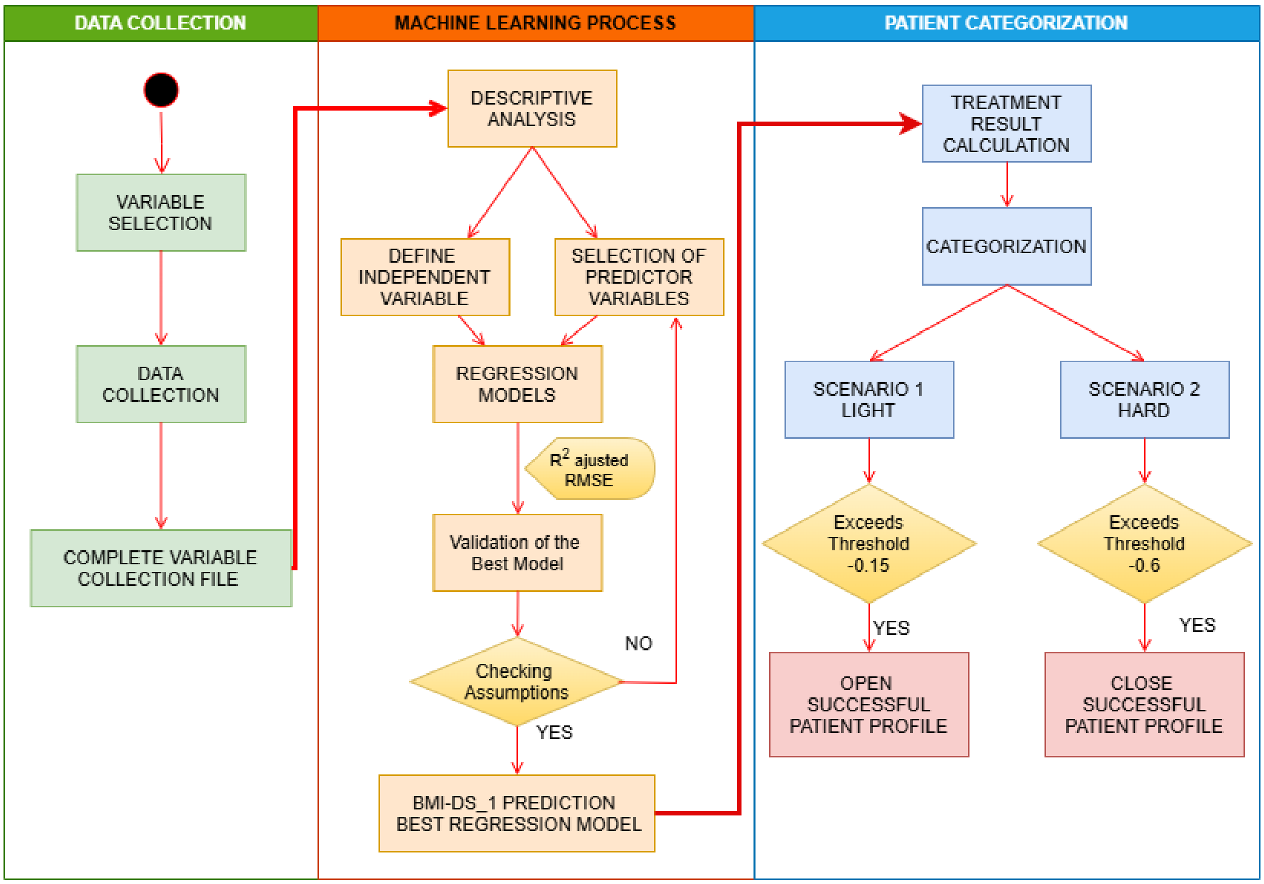


Figure 4: Summary diagram of the study process.

| Model | n_vars | R-sq | Adj R-sq | MSE | RMSE |
|---|---|---|---|---|---|
| Lineal Regression | 1 | 0.8050 | 0.8020 | 0.087339 | 0.295532 |
| Multiple Regression - (2) | 2 | 0.8080 | 0.8040 | 0.086645 | 0.294355 |
| Multiple Regression - (3) | 3 | 0.8080 | 0.8030 | 0.087586 | 0.295074 |
| ElasticNet | 3 | 0.8607 | 0.8500 | 0.058834 | 0.242558 |
| Lasso | 3 | 0.8607 | 0.8499 | 0.058837 | 0.242564 |
| Ridge | 3 | 0.8633 | 0.8528 | 0.057730 | 0.240270 |
| Polinomial | 3 | 0.8403 | 0.6966 | 0.059320 | 0.243521 |

Table 2: Comparison of regression models.

model. Finally, despite a good R-square of 0.8403, the polynomial model shows signs of overfitting with a much lower adjusted R-square of 0.6966 and an RMSE of 0.243521.

The Ridge regression model, which demonstrated the best performance with the lowest RMSE of 0.240270, utilizes three predictor variables. Equation 2 highlights the significant predictors and their coefficients, demonstrating the relationship between the dependent variable BMI-SD_1 and the predictor variables BMI-SD_0, t_v0_v1, and t_breastfeeding. The coefficients indicate the magnitude and direction of the relationship between each predictor and the outcome variable.

$$\text{BMI-SD\_1} = -0.29420 + (0.98340 \times \text{BMI-SD\_0}) + (0.01220 \times \text{t\_v0\_v1}) + (0.00270 \times \text{t\_breastfeeding}) \tag{2}$$

In order to refine the results further, several graphs have been created. Figures 5 and 6 are two violin plots showing the variable 'Result' distribution as a function of sex and age, respectively. These graphs were used to categorize the results and determine the importance of improving obesity treatment according to the two proposed scenarios.

In the analysis by gender, boys show more significant improvements than girls, achieving treatment success even in the most restrictive scenario. Regarding age analysis, younger children (5-10 years) have a higher success rate and significant improvements in obesity treatment than older children (11-15 years). As age increases, the proportion of these values decreases, suggesting that older children have less success and less significant improvements in treatment. In addition, children aged 14 and 15 show less variability, with almost all values above -0.15, suggesting that treatment success is very limited for these groups. On the other hand, there is a mix of results for children aged 6 to 9 years. Although many values are below -0.15, in-

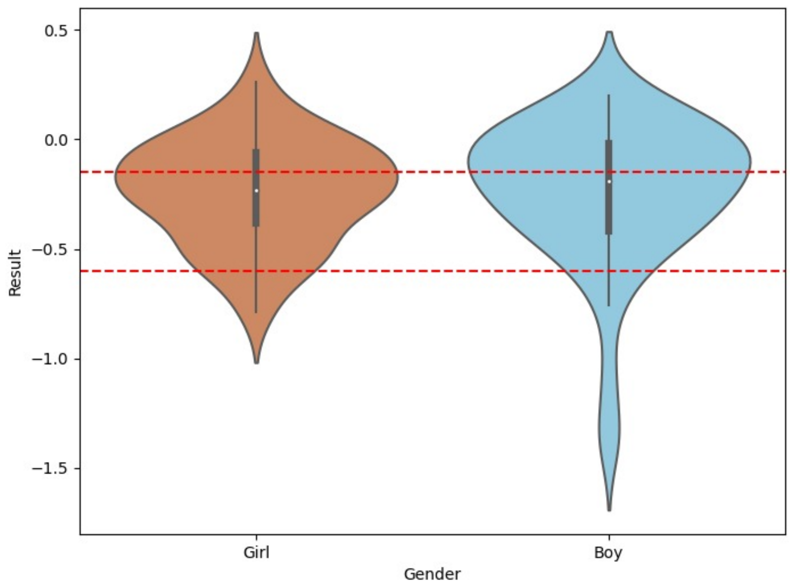


Figure 5: Violin distribution of the variable Result by gender.

dicating success, the variability suggests that not all individuals in these age groups have experienced the same level of improvement.

## 6. Discussion

Considering the various perspectives employed in categorizing the outcome, the variability in the results becomes evident, especially when considering the established criteria's flexibility. Based on these two scenarios, we will construct two distinct patient profiles, each oriented toward success according to the assumed criteria.

In the first scenario, where a laxer categorization criterion is adopted, patient profiles achieving success under more forgiving conditions will be identified. This approach will allow for capturing a broader spectrum of cases considered successful. In this case, a reduction of only -0.15 in BMI-SD_1 compared to the initial state of BMI-SD_0 is deemed sufficient to classify a case as successful. This generous threshold leads to more patients being classified in the success category, resulting in considerable variability in the outcomes.

When analyzing the correlation of the initial variables with BMI-SD_1 (Table 3), it is observed that, in this case, the result mainly depends on BMI-SD_0 and BMI_0. This indicates that the variability in treatment improvement is strongly influenced by the patient's initial conditions, focusing on the difference in BMI and initial BMI.

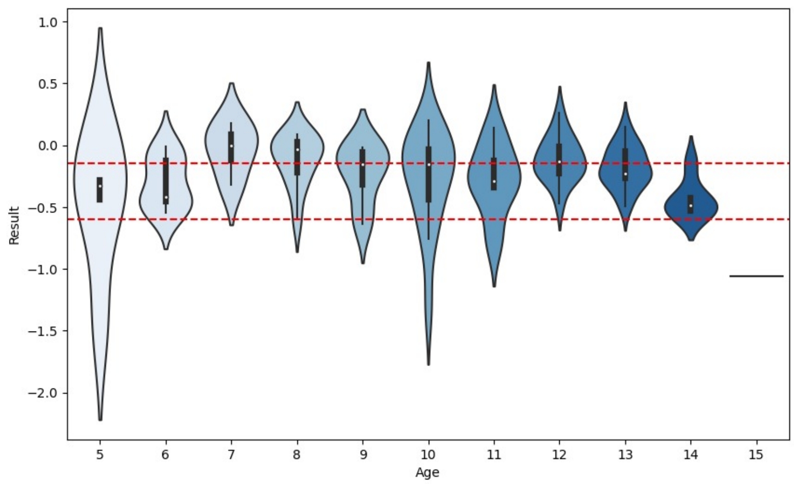


Figure 6: Violin distribution of the variable Result by age.

| Variable t=0 | Correlations BMI-SD_1 |
|---|---|
| BMI-SD_0 | 0.89401 |
| BMI_0 | 0.39314 |

Table 3: Notable correlations of variables with BMI-SD_1 in scenario 1.

This more lenient categorization approach highlights the importance of considering multiple factors in evaluating treatment success and underscores how specific initial parameters can determine factors in patient classification. Although few variables are directly associated with the final treatment outcome, Figure 7 shows the distribution of BMI-SD_0 and BMI_0 in successful cases, indicating that these cases are not very restrictive.

With these results, a considerable challenge arises in defining a distinctive profile for successful patients, as the variables used in Scenario 1 are not very restrictive, resulting in a broad set of cases classified as successful. The breadth of the definition of success under this generous criterion makes it challenging to identify specific characteristics that significantly distinguish successful patients from the rest.

On the other hand, in the second scenario, characterized by a more rigorous categorization criterion, a profile of patients will be delineated that are only considered successful under more strict and specific criteria. This more detailed and restrictive analysis aims to identify cases that meet higher standards of treatment success. Despite the modest proportion of patients (less than 10%) who achieved significant

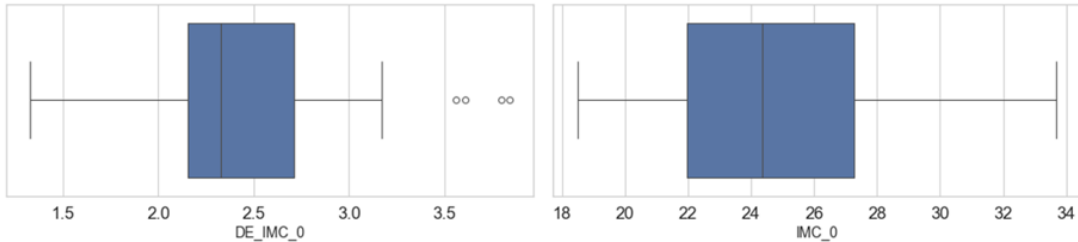


Figure 7: Distribution of values of important variables for successful patient in scenario 1.

improvements, several variables strongly correlate with "BMI-SD_1." Among them, "BMI-SD_0" exhibits a notable correlation, whose similarity to the correlation in the entire sample underscores its importance as a consistent predictor of BMI throughout the treatment.

Other variables emerge that until this point were not considered crucial in the context of the study (Table 4). Clinical variables such as "abdominal_girth," "hip_girth," "BMI_0," and "weight_0" reveal themselves as elements that could play a significant role in the positive evolution of patients. Variables related to habits, such as "KIDMED" and "sport," as well as the social variables "birth_country_mother" and "birth_country_father," also show notable connections with the patient's improvement. Including the genetic variable "height_father" highlights the inherent complexity of the interaction of various factors in the response to treatment.

In Figure 8, the profile of a patient who shows treatment success is defined, considering the critical variables for BMI-SD_1. This profile indicates that the successful patient generally presents an initial BMI-SD_0 value ranging between 2 and 3, classified as overweight on the BMI scale. Regarding dietary habits, it is highlighted that the patient follows a diet aligned with the KIDMED test. In addition, this patient is observed to participate in extracurricular sports activities, dedicating an average of 3 hours per week to exercise. From a medical perspective, the successful patient has an average BMI of around 24, an average abdominal circumference of 87 cm, and an average hip circumference of 87.5 cm.

These findings suggest that to significantly improve obesity treatment, the combination of factors such as starting treatment at overweight status, adherence to a balanced diet according to the KIDMED test, participation in regular sports activities and specific BMI values, abdominal circumference, and hip circumference could be determinant. These insights can be fundamental for personalizing and optimizing treatment strategies, allowing a more precise and effective approach for patients with similar profiles. These results demonstrate that adjusting the success thresholds can

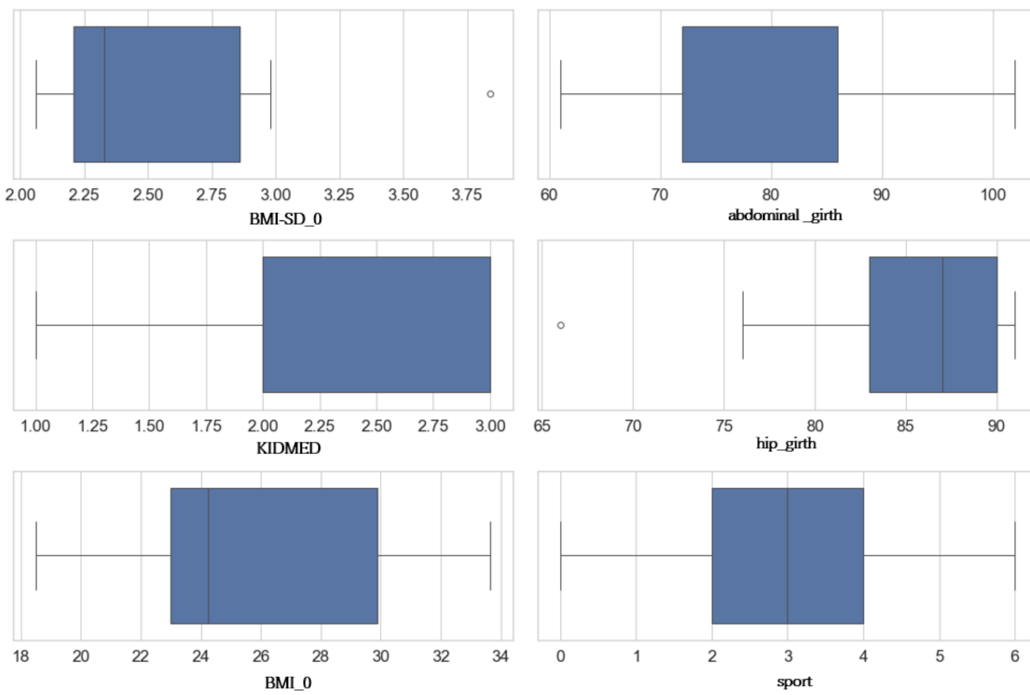


Figure 8: Distribution of values of important variables for successful patient in scenario 2.

| Variable t=0 | Correlations BMI-SD_1 |
|---|---|
| BMI-SD_0 | 0.851799 |
| abdominal_girth | 0.840261 |
| KIDMED | 0.773545 |
| hip_girth | 0.735447 |
| BMI_0 | 0.724276 |
| weight_0 | 0.567027 |
| birth_country_mother | 0.558155 |
| birth_country_father | 0.558155 |
| sport | 0.525808 |
| family_history | 0.504918 |
| size_0 | 0.466826 |
| t_visit0 | 0.365952 |
| height_father | 0.360211 |

Table 4: Notable correlations of variables with BMI-SD_1 in scenario 2.

be crucial in identifying key characteristics that influence treatment outcomes and help define a more accurate profile of successful patients.

## 7. Conclusion

In the context of this comprehensive analysis at the Elx-Crevillent Health Department, we have achieved a global understanding of childhood obesity in the population. Through the predictive regression model, key variables have been identified, highlighting the importance of the BMI-SD_0 as a fundamental predictor of treatment outcome and the study of the BMI standard deviation at the first post-treatment visit (BMI-SD_1) as the variable to be analyzed for treatment results. This emphasizes the importance of early starting the obesity interventions, as the benefits of early weight loss are more beneficial before puberty[2].

The Ridge regression model has proven robust and appropriate for this study due to its fit and simplicity. This model offers an effective tool for estimating BMI-SD_1, which is used to determine the outcome of the treatment. The results of this prediction can be crucial for specialists and facilitate clinical decision making and intervention strategies.

The detailed analysis of different classification thresholds for the outcome reveals several essential perspectives. With a lenient threshold, it is challenging to determine a successful patient profile, as the broad criterion allows for the inclusion of a large number of patients, making it difficult to identify specific characteristics. Conversely, with a stricter threshold, only a few patients meet the improvement standards. Medical variables, eating habits, and social factors are crucial to treatment success in this scenario. Adjusting the success thresholds is crucial, especially when considering factors such as starting treatment with overweight status, adherence to a balanced diet, and participation in sports activities.

The conclusions highlight the complexity of the factors involved in childhood obesity and emphasize the need for future research incorporating more complex variables, longitudinal measurements, and specific details about individual interventions. These recommendations aim to provide a deeper and more refined understanding, paving the way for more effective and personalized strategies in treating childhood obesity.

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